

Viacom18: Maximizing revenue and optimizing marketing campaigns for Voot VOD streaming with Google

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ABOUT VOOT

The Voot logo is a stylized, lowercase wordmark in a vibrant purple color. The letters are bold and rounded, with a slight shadow effect that gives it a three-dimensional appearance.

About Voot

With Google, Viacom18 is maturing the Voot streaming video-on-demand service to be a pre-evolved business marketing campaign budgets, and investments optimized to target

Owned by Viacom18, Voot is one of India's largest digital video-on-demand streaming services. With a content library of more than 65,000 hours across Viacom18's network content, exclusive content around network shows, licensed content, and Voot Originals, the platform caters to varying needs of discerning audiences across segments. The platform has about 100 million monthly active users.

Industries: Technology, Media & Entertainment

profitable custom
cohorts.

Location: India

Google Cloud results

- Optimizes crash analytics and identity and access management system cost and performance
- Creates pipeline to effectively capture customer data from multiple locations for analysis
- Achieves a 35% increase in customer retention

Identify
value
customer
cohorts
improve
marketing
campaigns
and ROI

Functions

Google Cloud (<https://cloud.google.com/>)

AutoML (<https://cloud.google.com/automl/>),

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://marketingplatform.google.com/about/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud AutoML (<https://cloud.google.com/automl/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about/>)

Google Ad Manager

(<https://admanager.google.com/home/>)

Google Analytics

(<https://marketingplatform.google.com/about/resources/analytics-product-overview/>)

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Firebase Crashlytics

(<https://firebase.google.com/docs/crashlytics/>)

India presents extensive opportunities for media and entertainment businesses, with the industry's value expected to reach \$25.56 billion in 2021–22 and the nation expected to become the world's most populous in the coming decades.

Entertainment network and brand owner Viacom18 Media Pvt Ltd (<https://www.viacom18.com/>) is ideally positioned to succeed in this fast-growing market. Founded in 2007 and now owned 51% by Network18 and 49% by ViacomCBS, Viacom18 Media Pvt Ltd styles itself as a provider of multi-brand and multi-generational experiences. In 2015 the business launched a digital division, Viacom18 Digital Ventures, to capitalize on the boom in over-the-top (OTT) streaming services and smartphone takeup in India.

Viacom18 Digital Ventures launched the Voot digital video-on-demand (VOD) streaming in May 2016, and consumer take up grew rapidly. The service recorded 50,000 monthly active users (MAUs) in its first month of operation, a number that grew to 100 million MAUs in less than five years. According to Viacom18, Voot is now India's second-largest premium digital VOD streaming service with a content library of more than 65,000 hours.

"Our digital platform features network content originally produced for television but made available as catch-up content, content produced through partnerships and licensing agreements with content creators in India and worldwide," says Anubhav Srivastava, Head of Data Science & Digital Ventures

at Viacom18. “We have also been producing critically acclaimed original content.”

Viacom18 Digital Ventures now operates three lines of business: the Voot advertising-supported app; Voot Kids, a learning app that provides ebook, video, audio and game content for children; and Voot Select, a premium subscriber version of the Voot app that features international shows and original content produced in India.

Anubhav is optimistic about the business’s growth potential in a market that saw 35% growth in digital in FY2020 due to an upsurge in the paid subscriber base across OTT platforms. However, like all players in the sector, Viacom18 faces challenges such as intense competition from a range of market entrants, including niche, small, and large businesses, as well as international operators—both those that have entered the market and those that have already established a presence in India.

In addition, Viacom18 must navigate changes to consumption habits and formats—highlighted by the short-form revolution sparked by video sharing social network TikTok, the evolution in subscription models, advances in ad formats, and the emergence of new monetization opportunities.

Voot a lucrative business with enormous potential

Viacom18 built the tools and systems that comprise Voot—including the player, login system, ad serving software, and content development network—on a

traditional cloud service. However, two years after launching Voot, Viacom18 realized it had a lucrative business with enormous potential. “We realized we were here to stay, compete, and do better,” says Anubhav. “This meant establishing a mindset of optimizing, improving, growing, and becoming more reliable. To do this while minimizing risk, we had to build our capabilities in-house.”

Viacom18 then began looking at products and vendors that could support this approach. “It made sense for us to deal with a provider like Google Cloud with an ecosystem that would allow us to develop our capabilities internally,” says Anubhav.

From a business perspective, Viacom18 wanted to unlock the potential that understanding and predicting user interactions with Voot could deliver, such as targeting campaigns more effectively or acquiring users from specific cohorts or demographics.

The business benchmarked Google Cloud against its incumbent cloud provider and other competitors and began using Google Cloud services for Voot on a case-by-case basis.

For example, Viacom18 determined Google Cloud was the easiest to integrate, lowest cost, easiest to scale, and most reliable for identity and access management and migrated its existing system across from a third-party vendor-managed service. In addition, after reviewing the cost and digital video recording performance of its incumbent third-party crash analytics tools, Viacom18 elected to move

them to Firebase Crashlytics

(<https://firebase.google.com/docs/crashlytics>), a lightweight, realtime crash reporting service.

Anubhav describes the service as “very, very reliable” with the flexibility to enable the organization’s development engineers to customize and build out solutions requested by its product teams. The performance and features of the service also prompted the business to move application performance management and indexing across to Firebase.

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—Anubhav Srivastava,
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Science|Digital Ventures,
Viacom18

Modeling customer LTV

However, the greatest rewards for Viacom18 and Voot lay in the ability to model the lifetime value (LTV) of its customers to understand which cohorts contributed the greatest financial value to Voot. However, disparate and siloed data challenged the creation of an effective LTV model.

“Most of our user information is stored in discrete locations based on its functional utility to the organization—for example, user behavior data resided in location A, marketing automation information in location B and advertising consumption behavior in location C,” explains Anubhav. “We found ascertaining lifetime value a difficult proposition, as we were working with multiple cloud vendors.”

A heavy user of Google services to manage its advertising and marketing campaigns, assets and spend, the business turned to the cloud provider to create a pipeline to move data from these sources to a single location for warehousing and analysis, from where it could develop use cases and solutions.

“Google came up with a compelling proposition to work jointly on the project with us,” says Anubhav.

“This initial step of creating a single location for all Voot data derived from different sources was the first step of the One Google approach.”

“Google provided some of the core components of the pipeline and a lot of on-ground support to enable us to push our data into a BigQuery (/bigquery) data warehouse for analysis,” says Anubhav. “With Google’s support, we completed the project in Q3 2019 and started training machine learning models in March 2020 using December 2019 data.”

A One Google approach delivers success

Google assigned expert design engineers and machine learning professionals to work with Viacom18 and Voot team members to create a pipeline to support the LTV project. “Google and our people used to jam every week and work together to define a project plan and work towards it,” says Anubhav. “The project involved input from a range of Google teams, including marketing, Ad Manager, and gTech [the Google Technical Services team]. With their help and support, we could run ‘intensive flow’ machine learning models on the data being fed into

BigQuery and apply that knowledge to our marketing automation activities.”

This One Google approach enabled the business to connect BigQuery to Google Cloud and machine learning, delivering a seamless pipeline from Google Cloud to Google Ads. “That was made possible through consultation from the gTech sell side, Cloud Security Engineering, and gTech buy side to run campaigns,” says Anubhav.

With data moving into BigQuery from a range of sources, Viacom18 has the history and connection to Voot touchpoints of every user. This allows the business to define key use cases such as the probability of users turning away from Voot or forecast incremental increases in viewership over a three to six month period.

These use cases are defined through the Google Cloud machine learning ecosystem—including AutoML (/automl), which allows Voot’s developers to easily create machine learning models, with the outputs pushed out to tools such as Google Analytics

(<https://marketingplatform.google.com/about/analytics/>)

and Google Ad Manager

(<https://admanager.google.com/home/>). The advertising exchange platform enables Viacom18 and Voot to manage user interactions with advertisements, run user acceptance testing for engagement or acquisition campaigns, and complement Campaign Manager 360, through which the business tracks conversions.

The machine learning algorithms enable Voot to build “lookalike” models of customers most likely to consume Voot content and fine-tune its user acquisition programs accordingly.

Working with Google has also given Voot the ability to develop and add new features quickly to the app through Cloud Functions (/functions) and change its appearance and behavior quickly with Firebase Remote Config.

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—Anubhav Srivastava,
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A 35% increase in customer retention

Early successes with the Google architecture include targeting marketing spend at users identified as most likely to turn away from the Voot app. “The customer cohort we targeted had a 35% higher rate of retention of the app relative to a control group,” says Rana Vishal Singh, Head of Growth Marketing at Voot. “We are continuing to identify where we can add value and direct our marketing dollars to obtain the best ROI. Like all organizations, we are on a constant quest to maximize the LTV of customers relative to the cost of customer acquisition through our campaigns.”

Efficiencies the business is targeting include understanding how frequently it should deliver advertisements—across video and digital formats—to maximize conversions. The business turned to Ads Data Hub to join Google’s event-level ad campaign data with its own data uploaded into BigQuery.

“Optimizing our customer acquisition campaigns solely on average numbers was not going to work, so we needed to apply automation through AI and machine learning,” says Anubhav. “Ads Data Hub enabled us to come up with a different, informed cost of acquisition appetite across all the customer cohorts we served, by helping us capture data about the advertising revenue derived from each cohort.”

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—Anubhav Srivastava,
Head of Data

LTV growing in importance

Connecting ad serving data with cost of acquisition data is becoming more and more important to Voot as the business scales. Anubhav and Rana Vishal point out that advertising revenues vary considerably between different customer cohorts, making identifying more lucrative groups essential to effective campaign strategies. Google has also helped Voot improve its performance in areas such as creating customized ad formats, enhancing marketing automation capabilities, and auditing to identify opportunities to reduce the application's size and improve its reliability. "Our sales, marketing, and advertising teams engage continuously with Google to improve Voot," explains Rana Vishal.

With Google products and services, and organic growth generated by its content decisions, Voot is now well positioned to operate as a profitable, evolved business at scale. "We realized we needed to be far more intelligent about our big marketing spends and return on investment to obtain meaningful value from each user, or be unit economics positive," says Rana Vishal.

"The use cases we've implemented and looking at for the future are creating significantly higher value for us.

“We are extremely confident in using Google’s capabilities to help us take this journey and, combined with our data and understanding of our users, they give us the ability to create a great future for Voot.”